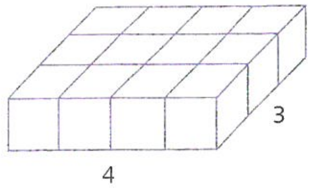
Slices and area



We can see that the rectangular prism is made up of five copies of the $4 \times 3 \times 1$ slice shown. The base area of the slice is 12 and the height is 1. We have 5 such slices in the original prism so we obtain volume $= 12 \times 5 = 60$ as before.

Thus the volume of a rectangular prism equals Ah, where A is the area of the base.

Note: The base can be any one of the three different faces of the prism.

The formulas above are still valid even when the dimensions are not whole numbers.

Right-rectangular prism

- A **right-rectangular prism** is a polyhedron in which:
  - the base is a rectangle
  - each cross-section parallel to the base is a rectangle congruent to the base
  - there are three right angles at each corner.
- Volume of a right-rectangular prism = length $\times$ width $\times$ height
  $= lwh$
- Volume of a right-rectangular prism = Area of base $\times$ height
  $= Ah$

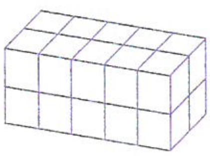
Exercise 15C

Example 7

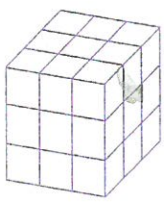
1 Find the volume of a rectangular prism whose dimensions are 12 mm, 10 mm and 7 mm.

2 If  is a 1-unit cube, find the volume of each of these figures in unit cubes.

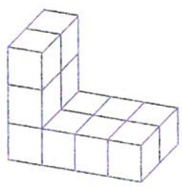
a



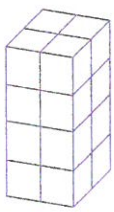
b



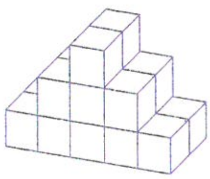
c



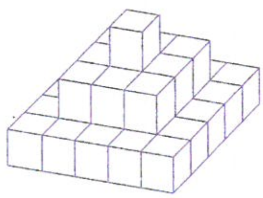
d



e



f



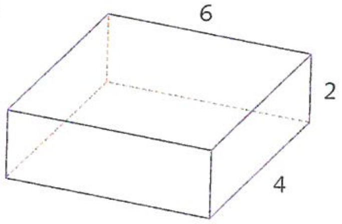
3 The table below gives the lengths, breadths, heights and volumes of various rectangular prisms. Fill in the missing entries.

	Length	Breadth	Height	Volume
a	3 cm	7 cm	6 cm	
b	5 cm	9 cm	4 cm	
c		3 m	2 m	18 m^3
d	3 m	12 m		144 m^3

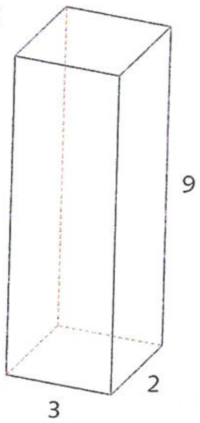
	Length	Breadth	Height	Volume
e	56 m	40 m	70 m	
f	2 mm	8 mm	16 mm	
g	32 m	4 m		1024 m^3
h	8 cm	8 cm		256 cm^3

4 Give the volume of each prism. All measurements are in cm.

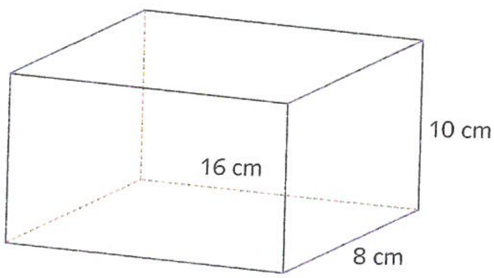
a



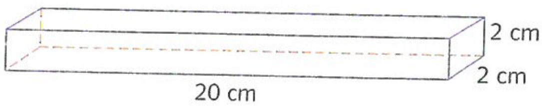
b



c



d

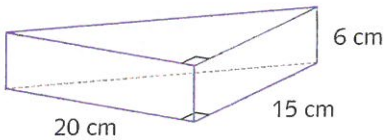


5 A swimming pool has dimensions 10 m by 12 m and is 5 m deep. What is its volume?

6 A rectangular water tank, whose base is 1.7 m by 1.4 m, holds water to a depth of 2 m. What is the volume of the tank, in cubic metres?

7 A Rubik's cube has side length 7 cm. What is its volume?

8 A piece of cheese sold in the shops is obtained by cutting a rectangular block of cheese in two. What is the volume of the cheese?



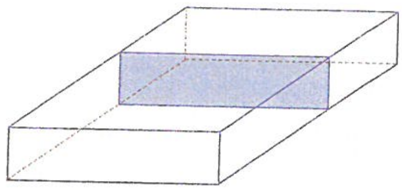
9 A tank in the form of a rectangular prism has base dimensions 2.3 m by 1.4 m and holds water to a depth of 2 m. If the depth is increased by 3 m, what is the increase in the volume of the water?

15D Volumes of other prisms

Recall that a **polyhedron** is a solid bounded by **polygons**. A **right prism** is a polyhedron that has two congruent and parallel faces called the base and top, and all its remaining faces are rectangles. A prism has **uniform cross-section**. This means that it is possible to take slices through the solid parallel to the base so that the area of each slice is always the same.

The name 'prism' comes from the Greek word *prizein*, which means 'to saw'. The idea is that if we 'saw' through a prism, the cross-sections are always the same.

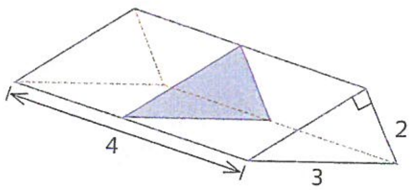
In a **rectangular prism**, the cross-section is always a rectangle.



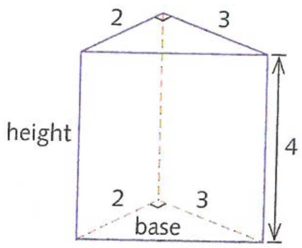
Volume of a triangular prism

In a **triangular prism**, the cross-section is always a triangle.

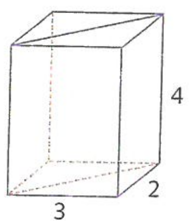
Here is a triangular prism whose cross-section is a right-angled triangle.



We can redraw the prism as shown here.



This prism can be thought of as the rectangular prism with dimensions $2 \times 3 \times 4$ cut in half.



$$\text{Volume of triangular prism} = \frac{1}{2} \times 2 \times 3 \times 4$$
$$= 12$$

In this case the base of the prism is the right-angled triangle with shorter side lengths 2 and 3.

The area of this triangle $= \frac{1}{2} \times 2 \times 3$.

The volume of the prism is given by the area of the base of the prism multiplied by the height.

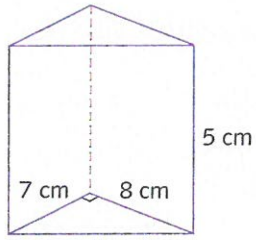
$$V = 3 \times 4$$
$$= 12$$

The volume of any triangular prism whose base has area A and whose height (or depth) is h is given by

$$\text{volume} = Ah$$

Example 8

Find the volume of the prism shown on the right.



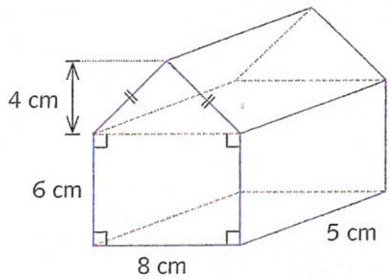
Solution

This is a triangular prism with area of base $A = \frac{1}{2} \times 8 \times 7$
$= 28 \text{ cm}^2$

$$\text{Volume} = Ah$$
$$= 28 \times 5$$
$$= 140 \text{ cm}^3$$

Example 9

Find the volume of the prism shown in the diagram.



Solution

The cross-section is the front face of the prism, a triangle on a rectangle.

$$A = \left(\frac{1}{2} \times 8 \times 4\right) + (8 \times 6)$$
$$= 64 \text{ cm}^2$$

$$\begin{aligned} \text{Volume} &= Ah \\ &= 64 \times 5 \\ &= 320 \text{ cm}^3 \end{aligned}$$

The base of the prism can be any polygon. Any polygonal prism can be be divided up into triangular prisms.

Volumes of other prisms

- A **right prism** is a polyhedron that has two congruent and parallel faces and all its remaining faces are rectangles.
- A prism has uniform cross-section.
- The volume of any prism whose base has area A and whose height is h is given by:

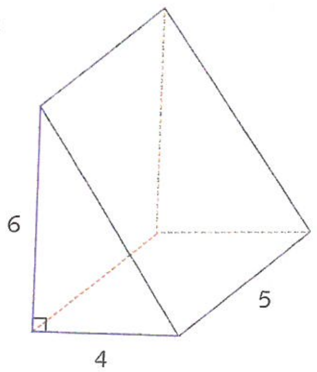
  volume of a prism $= Ah$

Exercise 15D

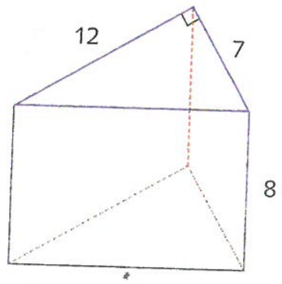
Examples 8, 9

1 Find the volume of each prism. All measurements are in centimetres.

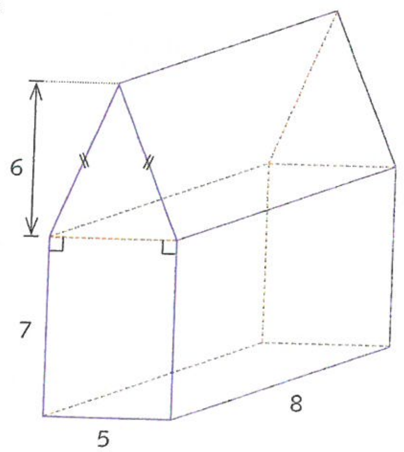
a



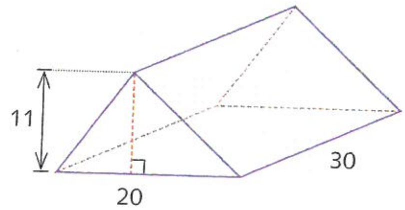
b



c



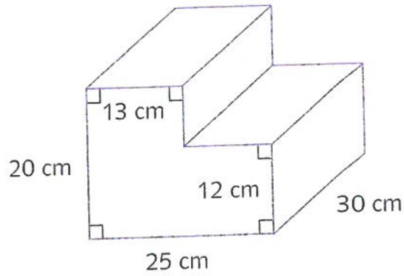
d



2 A small step has the shape shown opposite.

a Find the area of the front face.

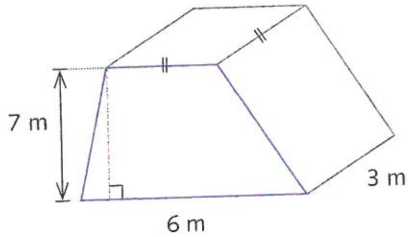
b Find the volume of the step.



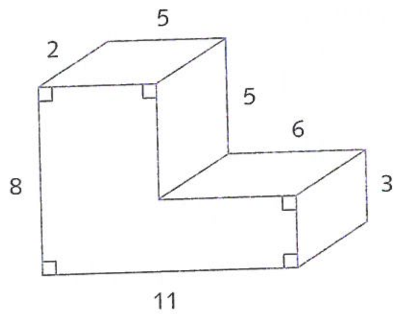
3 A large pedestal is in the shape of a prism whose front face is a trapezium.

a Find the area of the front face.

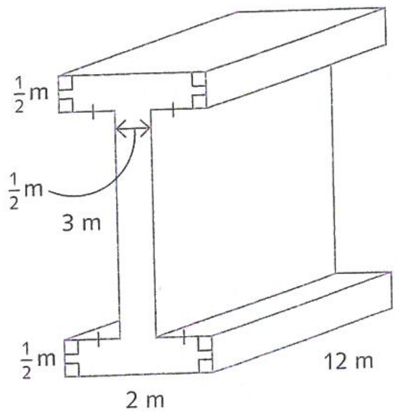
b Find the volume of the pedestal.



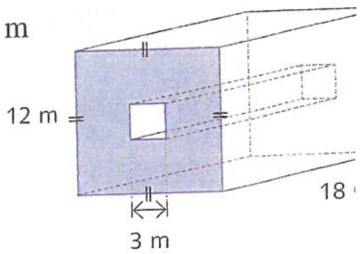
4 Find the volume of the stone block shown opposite, using the given measurements, which are in metres.



5 A steel girder has measurement in metres as shown. What is its volume?



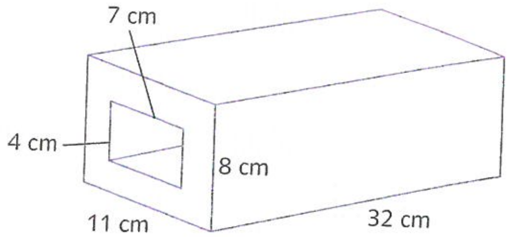
6 A 3 m by 3 m square prism is removed from a 12 m by 12 m square prism with depth 18 m. What is the volume of the remaining solid?



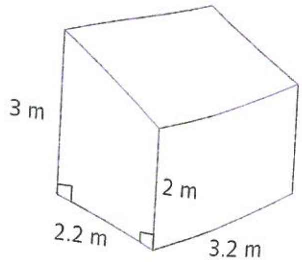
7 The front face of a prism is a parallelogram with base 8 cm and height 6 cm. If the depth is 12 cm, find the volume.

8 Find the volume of each prism.

a



b

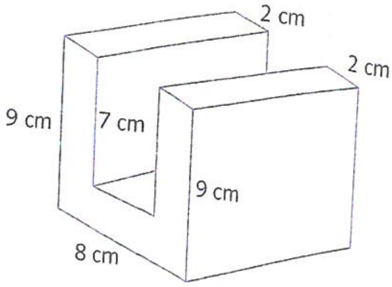


9 The volume of the prism is 792 cm^3.

Find:

a the area of the cross-section.

b the length of the prism.



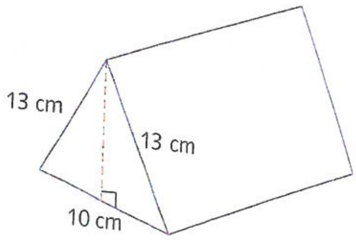
10 The volume of the prism is 4800 cm^3.

Find:

a the height of the triangle

b the area of the cross-section

c the length of the prism.



11 The cross-section of the prism is a trapezium of height 5 cm. The volume of the prism is 3900 cm^3. Find the length of the prism.